

From AI agents to humanoid robots

Top AI trends for 2025

—By **Jayesh Shinde**

From the emergence of autonomous AI agents capable of executing complex tasks to the integration of cutting-edge AI capabilities directly into devices, New Year 2025 promises to accelerate AI innovation at an unprecedented pace. The rise of humanoid robots, equipped with advanced AI systems, hints at a future where robots seamlessly coexist with humans, tackling labour-intensive tasks. Meanwhile, cost-efficient advancements in AI technologies aim to make these breakthroughs more accessible, addressing the challenges of high operational expenses.

Industry giants like OpenAI, Google, Microsoft, and NVIDIA are leading the charge, unveiling transformative technologies that will profoundly influence how we live and work. As these trends converge, 2025 is set to be a landmark year, ushering in a wave of AI-driven possibilities that will redefine our relationship with technology. Let's take a closer look...

If 2022 was the birth of AI chatbots as we know it, thanks to OpenAI's ChatGPT, then by all indications 2025 will see a lot of AI agents coming out into the open from their current secretive research bubble – and no I'm not talking about the garden variety agents referenced in *The Matrix*! How they will change our world is anyone's guess at this point, but the dawn of AI agents certainly promises to inject some excitement into the AI landscape that's becoming – dare I say – more drab by the day.

In the last two years, the world has seen a lot of breakneck advancement in the Generative AI space, right from text-to-text, text-to-image and text-to-video based Generative AI capabilities. And all of that's been nothing short of stepping stones for the next big AI breakthrough – AI agents.



In an interview with MIT Technology Review, OpenAI's Sam Altman described the best application of AI as a "super-competent colleague that knows absolutely everything about my whole life, every email, every conversation I've ever had, but doesn't feel like an extension." That's what an AI agent will look like, in essence. Are you ready for J.A.R.V.I.S., Tony Stark's intelligent AI assistant from *Iron Man*, or Samantha from *Her*, a significantly advanced AI operating system than what we've experienced till now?

Simply put, an AI agent is a slightly more advanced AI program that can perform certain autonomous tasks that aren't just limited to its own base program. ChatGPT or Gemini can write code for you, if you ask for it, but it can't go and create a website or an app from that code, where the website is live with a domain name or the app published on the app store. An AI agent will be able to do these things – I'm not saying these exact tasks that I suggested above, but these AI agents will have the ability to not just show what needs to be done but also go ahead and do some of that work.

According to Amazon's official AWS blog, humans will set broad goals for any given AI-related task, where the AI agent will independently choose the best actions it needs to perform to achieve those goals. Amazon further explains how in a customer service scenario, a future AI agent will automatically try to satisfy a calling customer's query – by looking up internal information, by asking different questions to the human

customer, by taking stock of the situation and responding with a solution that solves the calling customer's problem. In this scenario, the AI agent handles the customer's call on its own – without passing the call to a human customer support expert. In fact, whether or not to transfer a call to a human customer support expert is determined automatically by the AI agent. AI agents will be superior to simple AI chatbots thanks to their advanced reasoning capabilities, suggests IBM's blog post on AI agents. Unlike traditional AI chatbots like ChatGPT or Gemini, which give highly scripted responses to user queries, AI agents will have the ability to plan, think through and adapt to new information, enabling them to handle much more complex tasks with minimal human intervention or supervision.

All the big movers and shakers of the AI industry are planning to release their version of AI agents for the public very soon in 2025, if they haven't done it already by late 2024. OpenAI's reportedly working on getting their AI agent, codenamed Operator, out into the open for everyone to check out by January 2025. The AI agent is expected to be capable of autonomously operating certain tasks within your computer, like booking flight tickets and implementing code, among other things. **d**

